

Endpoint-lifted Schur and coherent causal grouping reduction

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-24 · this version issued 2026-07-24 · v1.0

Purpose and scope

This note continues the single live A13 child gate A13-CLASSII-NPC-CONE-MARTINGALE-INJECTION-BALANCE. It records one failed route and one repaired route without creating a peer-level claim or promoting the A13 tier. The failed route is the literal full-score affine tangent, which cannot be bounded uniformly in the derivative displacement because the production current matrix has rotating phase kernels. The repair evaluates the derivative-displacement column at the new endpoint. That endpoint lift removes the kernel defect exactly and gives a floor-uniform finite-dimensional Schur estimate.

The repaired local estimate is then inserted into a coherent frozen-value causal grouping. The control-derivative current telescopes pathwise, the interior mixed current telescopes after conditional expectation, and the only quadratic defect left by the local estimate contains the old Gaussian derivative, not the accumulated control derivative. This closes the local Schur part and the pure-control bookkeeping part of the R-068 successor.

Closed here. The affine full-score tangent is falsified by an exact production fixture. The endpoint-lifted current identity, optimized good/bad bound, global bound, coherent causal polarization, pathwise pure-control telescope, and fresh-noise mixed-current cancellation are proved at finite cutoff and uniformly in the positive density floor.

Still open. The adapted Gaussian-rooted transported-current/GG lower bound; the resulting full NPC-martingale injection balance; extension from regular one-shot controls to the declared finite-energy class; one-use; Nelson; and every floor, regulator, volume, and measure-construction limit.

All spatial integrations and sums over the three production generators and spatial directions are suppressed until Section 6. The local argument is pointwise. The fixed production matrix is $Q = Q_{\text{II}} > 0$, and

$$\lambda_Q = \|Q\|_{\text{op}} = \frac{219 + 3\sqrt{2129}}{16000P}, \quad P = 4 + 10^{-12}, \quad (0.1)$$

so $\lambda_Q = 0.005584738499072227$. Numerical values below are regression data derived from the pinned coefficients, not inputs to the proof. The density floor e in d_e is conceptually distinct from the mass denominator $P = M_X^2 + \text{classii_mass_regularizer}$ in Q_{II} , although the pinned production fixture uses 10^{-12} in both regularizers.

1 The current frame and its exact jets

Let S_A , $A = 1, 2, 3$, be the six-real symmetric Pauli generators. If P_D denotes the active-doublet projector, then $S_A^2 = P_D$. For a positive floor e put

$$d_e(z) = |z|^2 + e, \quad q_A(z) = \frac{z^T S_A z}{d_e(z)}, \quad r_A(z) = (S_A - q_A(z)I)z, \quad (1.1)$$

and introduce the two-column current frame

$$M_A(z) = [2S_A z, 2r_A(z)]. \quad (1.2)$$

For a derivative vector $y \in \mathbb{R}^6$, the corresponding two-current is $M_A(z)^T y$. Since

$$|r_A|^2 = |P_D z|^2 - q_A^2(|z|^2 + 2e) \leq |z|^2, \quad (1.3)$$

one has the floor-uniform frame bound

$$\boxed{\|M_A(z)\|_F \leq \sqrt{8}|z|}. \quad (1.4)$$

Direct differentiation gives

$$Dq_A(z)[h] = \frac{2r_A(z) \cdot h}{d_e(z)}, \quad (1.5)$$

$$D^2 q_A(z)[h, k] = \frac{2h \cdot (S_A - q_A I)k}{d_e} - \frac{4(r_A \cdot k)(z \cdot h)}{d_e^2} - \frac{4(r_A \cdot h)(z \cdot k)}{d_e^2}. \quad (1.6)$$

Writing $s_e(z) = \sqrt{d_e(z)}$, the elementary bounds $|q_A| \leq 1$, $|r_A| \leq |z| \leq s_e$, and $\|S_A - q_A I\|_{\text{op}} \leq 2$ yield

$$\boxed{\|DM_A(z)[h]\|_F \leq \sqrt{68}|h|, \quad \|D^2 M_A(z)[h, k]\|_F \leq \frac{32|h||k|}{s_e(z)}}. \quad (1.7)$$

Indeed $|Dq[h]| \leq 2|h|/s_e$, $|D^2 q[h, k]| \leq 12|h||k|/s_e^2$, and

$$D^2(qz)[h, k] = D^2 q[h, k]z + Dq[h]k + Dq[k]h, \quad (1.8)$$

whose norm is at most $16|h||k|/s_e$; only the second column of $D^2 M_A$ is nonzero and contributes the remaining factor two.

2 Why the affine full-score tangent fails

Set

$$B(z) = \sum_A M_A(z) Q M_A(z)^T, \quad f(z, y) = \frac{1}{2} y^T B(z) y. \quad (2.1)$$

For fixed z, a, y , the affine Taylor remainder in an arbitrary derivative displacement b contains

$$\frac{1}{2} b^T B(z+a) b + b^T [B(z+a) - B(z)] y. \quad (2.2)$$

Because $B(z+a)$ is merely positive semidefinite, (2.2) is bounded below in b if and only if

$$[B(z+a) - B(z)] y \in \text{Ran } B(z+a). \quad (2.3)$$

The production frames do not satisfy (2.3): they contain phase-kernel directions which rotate with z .

An exact six-real fixture makes the failure explicit. Take

$$z = (1, 0, 0, 0, 0, 0), \quad a = (0, \varepsilon, 0, 0, 0, 0), \quad y = (0, 0, 0, 0, 1, 0), \quad (2.4)$$

and choose the new-endpoint vertical direction

$$b = t(0, 0, 0, 1, \varepsilon, 0). \quad (2.5)$$

For every positive floor, $M_A(z + a)^T(y + b) = M_A(z)^T y$ for all A . Consequently the raw secant is zero, whereas the old affine tangent equals

$$\mathcal{T}_{\text{aff}} = \beta_0 \varepsilon t, \quad \beta_0 = 4(a_{\text{prod}} + 2b_{\text{prod}} + c_{\text{prod}}) = \frac{339}{2000P} > 0. \quad (2.6)$$

Thus the affine remainder is $-\beta_0 \varepsilon t$ and has no uniform lower bound. At $\varepsilon = 0.1$ and $t = 7$, the exact values are

$$\mathcal{R} = 0, \quad \mathcal{T}_{\text{aff}} = 0.0296625, \quad \mathcal{R} - \mathcal{T}_{\text{aff}} = -0.0296625. \quad (2.7)$$

This falsifies only the affine-tangent architecture. It is not a counterexample to the raw secant theorem or to the endpoint-lifted repair.

3 The endpoint-lifted identity

Define the old current, exact increment, endpoint-lifted linear part, and frame-curvature remainder by

$$w_A = M_A(z)^T y, \quad (3.1)$$

$$\delta_A = M_A(z + a)^T(y + b) - M_A(z)^T y, \quad (3.2)$$

$$L_A = DM_A(z)[a]^T y + M_A(z + a)^T b, \quad (3.3)$$

$$E_A = [M_A(z + a) - M_A(z) - DM_A(z)[a]]^T y. \quad (3.4)$$

Then

$$\boxed{\delta_A = L_A + E_A.} \quad (3.5)$$

The crucial point is that E_A is independent of b . With $\langle u, Qv \rangle_A = \sum_A u_A^T Q v_A$ and $|u|_Q^2 = \langle u, Qu \rangle_A$, the raw current secant is

$$\mathcal{R} = \langle w, Q\delta \rangle_A + \frac{1}{2}|\delta|_Q^2. \quad (3.6)$$

Therefore, exactly,

$$\boxed{\mathcal{R} = \mathcal{T}_{\text{end}} + \mathcal{J}_{\text{end}} + \sum_A \langle w_A, QE_A \rangle,} \quad (3.7)$$

where

$$\mathcal{T}_{\text{end}} = \langle w, QL \rangle_A, \quad \mathcal{J}_{\text{end}} = \frac{1}{2}|\delta|_Q^2 \geq 0. \quad (3.8)$$

The object L is a hybrid endpoint lift, not the Fréchet derivative at the old endpoint: its a column is differentiated at z , while its b column is evaluated at $z + a$. This precise asymmetry is what removes the rotating- kernel obstruction. The full exact Jacobi square is retained; no completion divides by a possibly singular current matrix.

4 Optimized good/bad and global Schur bounds

Fix $0 < \theta < 1$. In the good region $|a| \leq \theta s_e(z)$, every point of the segment satisfies

$$s_e(z + ta) \geq s_e(z) - t|a| \geq (1 - \theta)s_e(z). \quad (4.1)$$

Taylor's integral formula and (1.7) give

$$\|M_A(z + a) - M_A(z) - DM_A(z)[a]\|_F \leq \frac{16|a|^2}{(1 - \theta)s_e(z)}. \quad (4.2)$$

Combining (4.2) with (1.4) and summing the three generators gives

$$\left| \sum_A \langle w_A, QE_A \rangle \right| \leq \frac{48\sqrt{8}\lambda_Q}{1 - \theta} |a|^2 |y|^2. \quad (4.3)$$

In the bad region $|a| > \theta s_e(z)$, positivity of the new current square in (3.6) gives

$$\mathcal{R} \geq -\frac{1}{2}|w|_Q^2 \geq -12\lambda_Q|z|^2|y|^2 \geq -\frac{12\lambda_Q}{\theta^2}|a|^2|y|^2. \quad (4.4)$$

Hence the resulting literal good/bad statement is

$$\boxed{\mathcal{R} \geq \mathbf{1}_{\{|a| \leq \theta s_e(z)\}} (\mathcal{T}_{\text{end}} + \mathcal{J}_{\text{end}}) - C_\theta |a|^2 |y|^2,} \quad (4.5)$$

$$C_\theta = \max \left\{ \frac{48\sqrt{8}\lambda_Q}{1 - \theta}, \frac{12\lambda_Q}{\theta^2} \right\}. \quad (4.6)$$

Optimizing these two safe envelopes gives (not a claim of global sharpness)

$$\theta_\# = \frac{2}{1 + \sqrt{1 + 32\sqrt{2}}} = 0.25637440844670695, \quad (4.7)$$

$$\boxed{C_\# = 3\lambda_Q(1 + \sqrt{1 + 32\sqrt{2}})^2 = 1.0196115487028556.} \quad (4.8)$$

There is also a weaker but globally endpoint-lifted form. The global Lipschitz bound in (1.7) implies $\|E_A\| \leq 2\sqrt{68}|a||y|$. On the bad region, $|z| < |a|/\theta$, so

$$\left| \sum_A \langle w_A, QE_A \rangle \right| \leq \frac{24\sqrt{34}\lambda_Q}{\theta} |a|^2 |y|^2. \quad (4.9)$$

Balancing (4.3) with (4.9) gives

$$\theta_* = \frac{\sqrt{34}}{\sqrt{34} + 4\sqrt{2}} = 0.5075774975293578, \quad (4.10)$$

and the floor- and b -uniform inequality

$$\boxed{\mathcal{R} \geq \mathcal{T}_{\text{end}} + \mathcal{J}_{\text{end}} - C_* |a|^2 |y|^2, \quad C_* = 24\lambda_Q(\sqrt{34} + 4\sqrt{2}) = 1.5397534378598672.} \quad (4.11)$$

5 Coherent frozen-value causal grouping

For a terminal shell J , couple only the heat-dummy values by

$$r_{j:J} = \sum_{k=j}^J \xi_k, \quad U = X_{<J} + r_{j:J} = \sum_{k \leq J} \xi_k. \quad (5.1)$$

Thus U is the same value variable at every shell. Equation (5.1) is a target-space heat coupling: $r_{j:J}$ is never spatially differentiated and must not be identified with the actual future spatial field. This is exactly the backward-heat convention proved in R-066.

Write

$$z_j = U + A_{<j}, \quad z_{j+1} = z_j + a_j, \quad (5.2)$$

$$G_{j,i} = D_i X_{<j}, \quad C_{j,i} = D_i A_{<j}, \quad C_{j+1,i} = C_{j,i} + b_{j,i}. \quad (5.3)$$

For every generator and spatial direction suppress the indices and set

$$c_j = M(z_j)^T C_j, \quad g_j = M(z_j)^T G_j, \quad (5.4)$$

$$c_{j+1} = M(z_{j+1})^T (C_j + b_j), \quad g_j^+ = M(z_{j+1})^T G_j. \quad (5.5)$$

Let $\delta_j^C = c_{j+1} - c_j$ and $\delta_j^G = g_j^+ - g_j$. Polarization gives the exact split

$$\mathcal{R}_j = \mathcal{R}_j^C + \mathcal{R}_j^G + \mathcal{R}_j^\times, \quad (5.6)$$

$$\mathcal{R}_j^C = \frac{1}{2}(|c_{j+1}|_Q^2 - |c_j|_Q^2), \quad (5.7)$$

$$\mathcal{R}_j^G = \frac{1}{2}(|g_j^+|_Q^2 - |g_j|_Q^2), \quad (5.8)$$

$$\begin{aligned} \mathcal{R}_j^\times &= \langle c_j, Q\delta_j^G \rangle + \langle g_j, Q\delta_j^C \rangle + \langle \delta_j^C, Q\delta_j^G \rangle \\ &= \langle c_{j+1}, Qg_j^+ \rangle - \langle c_j, Qg_j \rangle. \end{aligned} \quad (5.9)$$

In particular, the pure-control current creation telescopes pathwise:

$$\boxed{\sum_{j=j_0}^J \mathcal{R}_j^C = \frac{1}{2}|c_{J+1}|_Q^2 - \frac{1}{2}|c_{j_0}|_Q^2.} \quad (5.10)$$

For an empty initial high-block control the right-hand side is nonnegative; otherwise only the fixed low boundary remains.

If $G_{j+1} = G_j + d_j$ and $n_j = M(z_{j+1})^T d_j$, then

$$g_{j+1} = g_j^+ + n_j. \quad (5.11)$$

Strict-past adaptation makes the control pre-shell, while c_{j+1} may still depend on the current value ξ_j through z_{j+1} . It is, however, pre-derivative-increment. For the common real-even regulator, the same-point value ξ_j and derivative increment d_j are independent, so

$$\mathbb{E}_j \langle c_{j+1}, Qn_j \rangle = 0. \quad (5.12)$$

Thus the interior terms of (5.9) telescope in expectation. The terminal mixed current is not discarded: it must remain coupled to the terminal $CC + GG$ terms as the full terminal current square.

For completeness, the exact Gaussian companion identity is

$$\mathcal{R}_j^G = \frac{1}{2}(|g_{j+1}|_Q^2 - |g_j|_Q^2) - \langle g_j^+, Qn_j \rangle - \frac{1}{2}|n_j|_Q^2, \quad (5.13)$$

and the same conditional independence gives

$$\mathbb{E}_j \langle g_j^+, Qn_j \rangle = 0. \quad (5.14)$$

Thus the exact, unreplaced GG sum has the usual terminal-square minus fresh-injection telescope. Together with (5.10) and (5.9), its terminal three terms form $|c_{J+1} + g_{J+1}|_Q^2/2$. This exact identity is logically separate from Section 6, where $\mathcal{R}_j^G$ is replaced by a lower bound.

This cancellation is pointwise/marginal in the target-space heat coupling and expectation-only. Analytically it uses whole-shell $\mathcal{F}_{j-1}$ -measurability, integrability/Fubini, and the pinned identity $\mathbb{E}[\xi_j(x) \otimes d_{j,i}(x)] = 0$. It is neither a pathwise cancellation nor a license for progressive within-shell feedback. If summation begins after a finite-low split, the combined exact telescope retains the full initial boundary $-|c_{j_0} + g_{j_0}|_Q^2/2$ until the existing finite-low estimate absorbs it.

6 What the repaired lemma buys

Apply (4.11) only to $\mathcal{R}_j^G$, with $y = G_j$ and $b = 0$. Define

$$E_j^G = [M(z_j + a_j) - M(z_j) - DM(z_j)[a_j]]^T G_j, \quad (6.1)$$

$$L_j^G = DM(z_j)[a_j]^T G_j, \quad \delta_j^G = L_j^G + E_j^G. \quad (6.2)$$

Then

$$\boxed{\mathcal{R}_j^G \geq \mathcal{T}_j^G + \mathcal{J}_j^G - C_* |a_j|^2 |G_j|^2}, \quad (6.3)$$

where

$$\mathcal{T}_j^G = \langle g_j, QL_j^G \rangle, \quad \mathcal{J}_j^G = \frac{1}{2} |\delta_j^G|_Q^2. \quad (6.4)$$

The important structural fact is literal: (6.3) has no factor $|C_j|^2$. The dangerous defect

$$\sum_j |a_j|^2 |DA_{<j}|^2 \quad (6.5)$$

has disappeared because its underlying CC energy was telescoped exactly in (5.10).

After normal ordering, the GG coefficient secant is

$$\frac{1}{2} [B(z_{j+1}) - B(z_j)] : (G_j \otimes G_j - \Gamma_{j-1}). \quad (6.6)$$

The transfer of the centered scalar defect to R-068 is an expectation identity, not a pathwise identification with a spatial projection. Sum over the suppressed spatial directions and trace the six-real field indices:

$$\tilde{q}_J = \sum_i (|G_{J+1,i}|^2 - \text{Tr } \Gamma_{J,i}), \quad \tilde{q}^{<j} = \sum_i (|G_{j,i}|^2 - \text{Tr } \Gamma_{j-1,i}). \quad (6.6a)$$

Independence of future derivative shells gives $\tilde{q}^{<j} = \mathbb{E}[\tilde{q}_J \mid \mathcal{F}_{j-1}]$. Since $|a_j|^2$ is $\mathcal{F}_{j-1}$ -measurable,

$$\mathbb{E} \sum_j \langle \tilde{q}^{<j}, |a_j|^2 \rangle = \mathbb{E} \langle \tilde{q}_J, \Pi_0(a) \rangle, \quad \Pi_0(a) = \sum_j |a_j|^2. \quad (6.6b)$$

If $A = (\sum_j |a_j|^2)^{1/2}$, $X = \sum_j \|a_j\|_{H^2}^2$, and $Y = \|A\|_6^6$, then

$$\|\Pi_0(a)\|_2 \leq \|A\|_4^2 \leq C_L Y^{1/3}, \quad \|\Pi_0(a)\|_{H^2} \leq C_H X. \quad (6.6c)$$

Interpolation and the same three-factor Young inequality as R-068 therefore pay (6.6b) once from the terminal $H^{-1-\kappa}$ model norm. The covariance trace costs $O(N_j^{-3}) \|h_j\|_{\mathcal{H}_j}^2$. This argument applies to the

displayed quadratic defect only. It does not make every expression containing a Gaussian derivative eligible for the uncontrolled N^{-1}/N^{-2} score tails.

Combining (5.10), (5.12), and (6.3) reduces the remaining load-bearing statement to one adapted Gaussian-rooted transported-current estimate. In particular, the terminal control-current square may not be dropped:

$$\mathbb{E} \sum_{j=j_0}^J (\mathcal{T}_j^G + \mathcal{J}_j^G + \mathcal{R}_j^\times) + \frac{1}{2} \mathbb{E}(|c_{J+1}|_Q^2 - |c_{j_0}|_Q^2) \geq -\eta \mathbb{E} \sum_j \|h_j\|_{\mathcal{H}_j}^2 - \zeta \mathbb{E} \|Z_J\|_6^6 - C_{\eta, \zeta, j_0}. \quad (6.7)$$

The coefficient in (6.7) depends on the adapted past control. Existing Gaussian tails cannot be cited until (6.7) is reduced pathwise to the pinned global centered form plus its covariance trace. The control-current boundary in (6.7) cannot be omitted because the mixed term by itself has no lower bound. Eventual terminal bookkeeping may recombine it with the corresponding mixed and Gaussian boundaries, but no such additional telescope is claimed here. Equation (6.7), with the exact control boundary and intact mixed secant, is a sufficient next analytic target.

7 Why the pure-control defect cannot be paid separately

There is also a scaling reason not to return to (6.5). On the normalized circle take two shells

$$a_N = tN \cos Nx, \quad a_{2N} = tN \cos 2Nx. \quad (7.1)$$

At the second shell,

$$\int |a_{2N}|^2 |Da_N|^2 = \frac{1}{4} t^4 N^6, \quad (7.2)$$

whereas the leading inhomogeneous H^2 cost and terminal sextic cost are

$$\sum \|a_j\|_{H^2}^2 = \frac{17}{2} t^2 N^6 + O(N^4), \quad \|a_N + a_{2N}\|_6^6 = \frac{215}{32} t^6 N^6. \quad (7.3)$$

All three terms have the same cutoff power. For arbitrarily small prescribed η and ζ , the coefficient

$$\frac{t^4}{4} - \frac{17\eta t^2}{2} - \frac{215\zeta t^6}{32} \quad (7.4)$$

is positive for a suitable fixed t ; at $t = 1$ and $\eta = \zeta = 10^{-5}$ it is 0.2498478125. Hence a direct separate estimate of (6.5) cannot meet the required arbitrary budgets uniformly in N . This is an architecture no-go, not a counterexample to the exact telescope.

8 Independent execution and proof boundary

The primary executable derives Q , every displayed constant, the exact frame jets, the endpoint identity, both Schur bounds, the rotating-kernel fixture, the coherent polarization, ninth-order Gauss–Hermite fresh-noise centering, and the two-shell scaling fixture. It passes 22/22 assertions. The non-importing executable reconstructs the Pauli frames and production coefficients, uses a different random seed and eleventh-order Hermite rule, checks centered finite-difference second-order convergence, and passes 20/20 assertions. In the independent audit, halving the centered-difference step reduces the worst frame-jet error from 6.61075×10^{-7} to 1.72164×10^{-7} .

The Gauss–Hermite fixtures illustrate centering under an independent value/derivative product law. The proof of (5.12) is instead the analytic strict-past argument with the pinned common real-even regulator and the integrability assumptions stated after (5.12).

These finite-dimensional executions are regression and falsification evidence. They do not replace the analytic estimates (1.4)–(7.4), and they do not test arbitrary adapted fields. The fail-closed package command is

9 Devil’s-advocate review

1. **“The old affine tangent is uniformly valid because the Jacobi term is positive.” UPHELD as false.** The Schur quadratic is bounded below only under the range condition (2.3). The rotating-kernel fixture violates it and makes the affine remainder linear and unbounded in the vertical amplitude.
2. **“The endpoint lift merely hides the same kernel problem.” DISMISSED.** Equations (3.3)–(3.5) remove b from the remainder algebraically before estimation. Bounds (4.5) and (4.11) are consequently uniform in b , including vertical kernel directions.
3. **“The future heat dummy may be differentiated as a future spatial field.” UPHELD as false.** The coupling (5.1) is only a target-space value coupling. Differentiating $r_{j:J}$ would add a future derivative not present in the backward-heat identity.
4. **“Every term with a Gaussian root inherits the R-068 N^{-1}/N^{-2} tails.” UPHELD as false.** Those tails are proved at the uncontrolled Gaussian base. The adapted coefficient in (6.7) is the remaining theorem, not an already centered test function.
5. **“The mixed current can be discarded after its interior increments center.” UPHELD as false.** Only the fresh-noise interior correction in (5.12) centers. The terminal cross must remain with the terminal $CC + GG$ square; separating it can create a negative $-G^2$ term even in a flat scalar diagnostic.
6. **“The pure-control defect can simply be absorbed by H^2 and L^6 .” UPHELD as false.** Equations (7.1)–(7.4) put all three quantities at order N^6 and leave a positive coefficient for small arbitrary budgets. The exact telescope (5.10) is essential.
7. **“This reduction proves controlled-shell one-use.” UPHELD as false.** The adapted lower bound (6.7), finite-energy extension, and downstream one-use/Nelson steps remain open. No tier change is justified.

Result footer

Result ID:	A13-CLASSII-ENDPOINT-LIFTED-SCHUR-CAUSAL-GROUPING-REDUCTION
Precise statement:	The hybrid endpoint lift removes derivative displacement from the curvature remainder and gives optimized floor-uniform good/bad and global bounds; coherent value grouping telescopes pure-control and fresh derivative-noise currents in their stated sense.
Scope:	Fixed finite cutoff; every positive floor; regular strict-past one-shot shells; target-space heat values.
Dependencies:	A1 production functional; R-066; R-067; R-068.
Evidence grade:	ANALYTIC, EXACT, EXECUTED.
Reproduction command:	python codes/foundations/ a13_classii_endpoint_lifted_schur_causal_grouping_ reduction_verify.py
Expected output:	primary 22/22; independent 20/20; integrated 70/70; aggregate 112/112; exit 0.
Falsification gate:	Endpoint identity or constants fail; a causal telescope, source/PDF hash, PDF QA, or assertion fails.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	The adapted transported-current/GG bound, arbitrary finite-energy drift, one-use, Nelson, measure, and floor/regulator/volume limits remain open.
Next required action:	Prove (6.7) with the terminal current square and mixed secant retained, then make the finite-energy extension.

Final boundary. The local production Schur estimate and the pure-control causal bookkeeping are no longer the obstruction. The exact remaining bottleneck is (6.7): an adapted Gaussian-rooted transported-current/GG lower bound which preserves the mixed secant and reduces its genuinely centered part to the existing R-068 global quadratic-form theorem.